

February 22, 2025

Mr. Steven Posnack, MS, MHS
Principal Deputy Assistant Secretary for Technology Policy
Principal Deputy National Coordinator for Health Information Technology
The Office of the National Coordinator for Health IT
Department of Health and Human Services
Hubert Humphrey Building, Suite 729
200 Independence Avenue SW, Washington, DC 20201

Submitted electronically to:

<https://www.regulations.gov/commenton/HHS-ONC-2026-0001-0001>

Re: Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

Dear Mr. Posnack:

The Health Data Standards division of the Regenstrief Institute appreciates the opportunity to respond to the Assistant Secretary for Technology Policy's Request for Information on accelerating the adoption and use of artificial intelligence in clinical care. As a research organization with deep expertise in biomedical informatics, learning health systems, and national health data standards stewardship, Regenstrief approaches AI not as a standalone technology, but as a capability that depends on strong data foundations, real-world evidence, and sustained governance.

Our responses focus on areas where federal leadership can most effectively translate AI innovation into durable clinical impact. In particular, we emphasize the importance of semantic and technical interoperability, standards-based data infrastructure, pragmatic evaluation in real-world settings, and post-deployment monitoring to support safe, trustworthy, and scalable use of AI in care delivery.

The enclosed responses address three priority areas:

- Key barriers to private sector innovation and adoption of AI in clinical care
- The role of enhanced interoperability in widening markets, fueling research, and enabling generalizable AI evidence
- Patient and caregiver priorities and concerns related to AI use in clinical settings

Across these areas, our comments reflect findings from the published literature and practical experience supporting national health data standards, multi-site research, and learning health system infrastructure. We believe sustained progress in clinical AI adoption will depend less on algorithmic advances alone and more on coordinated investment in data standards,

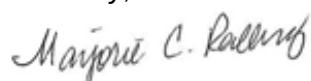
interoperability, real-world evidence generation, and governance mechanisms that enable responsible use over time.

We appreciate ASTP's leadership in convening stakeholders across government, industry, and the research community, and we welcome continued engagement on policies and programs that support trustworthy, evidence-based integration of AI into clinical care.

If there are questions regarding our comments, please contact Marjorie Rallins, DPM, MS, Executive Director, Health Data Standards at mrallins@regenstrief.org or 317-274-9415.

We appreciate and look forward to continued collaboration with ASTP.

Sincerely,



Marjorie Rallins, DPM, MS
Executive Director
Regenstrief Institute Health Data Standards

Enclosure: Regenstrief Institute Health Data Standards Response Summary

Regenstrief Institute Response Summary

Office of the National Coordinator for Health Information Technology (ASTP)

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Specific Questions

Question 1: What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

Comment: At their core, many barriers to AI adoption stem from inconsistent, non-standardized clinical data and limited semantic interoperability, which undermine technical performance and trust. AI systems depend on computable, comparable data across settings, yet uneven adoption and alignment of foundational standards—such as LOINC®, UCUM, SNOMED CT®, and others—continue to require costly data normalization and limit the ability to validate, scale, and monitor AI tools in real-world clinical environments. These gaps contribute to clinician skepticism and limit the ability to generate transferable, real-world evidence across institutions. Strengthening national adoption and coordinated use of core clinical data standards would lower development and implementation costs, improve model reliability and transparency, and support more robust evaluation of safety and effectiveness. Addressing these data foundations, alongside regulatory clarity and payment alignment, is essential to building confidence in AI and unlocking its full potential in clinical care.

Question 8: Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

Comment: Enhanced interoperability would most effectively widen market opportunities and accelerate AI development where it enables semantically consistent, standards-based, and longitudinal clinical data that can be reused across organizations, vendors, and care settings.

1. Foundational Clinical Data Prioritized in USCDI

AI development and deployment depend on consistent representation of core clinical data prioritized within USCDI, including laboratory results, vital signs, clinical observations, diagnoses, procedures, medications, and outcomes. Stronger interoperability across these data types—through coordinated use of nationally recognized data, terminology, and units of measure standards—would reduce data normalization costs, improve model generalizability, and expand market access for AI developers.

2. Semantic and Technical Interoperability Across the Standards Ecosystem

True interoperability requires shared clinical meaning and reliable technical exchange. Uneven adoption and misalignment across clinical terminologies, information models,

exchange specifications, and implementation guides continue to limit AI development, evaluation, and deployment. Greater coordination across the standards ecosystem would enable AI models trained in one environment to be more readily validated, deployed, and monitored in others without extensive reengineering.

3. Longitudinal and Cross-Setting Data

AI tools that support prediction, risk stratification, quality measurement, and outcomes analysis require longitudinal data spanning inpatient, outpatient, laboratory, public health, and community-based care settings. Interoperability that supports consistent use of standards-aligned data over time and across settings would fuel research and strengthen real-world evidence generation.

4. Benchmarking, Evaluation, and Real-World Evidence Infrastructure

Enhanced interoperability is essential for shared benchmarking tools, reference datasets, and evaluation frameworks. When AI inputs and outputs are grounded in common standards, performance metrics can be compared meaningfully across vendors and institutions, supporting post-deployment monitoring and adoption.

5. Supporting Evolution and Innovation in Standards

Interoperability frameworks should support responsible integration of emerging data types and evolving standards while maintaining backward compatibility and alignment with USCDI.

Question 9: What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

Comment: Research consistently shows that patients and caregivers support AI adoption when it improves access, coordination, safety, and personalization of care while preserving trust and the clinician–patient relationship. Patients value AI applications that improve timeliness of care, reduce diagnostic delays, support care coordination for complex conditions, reduce documentation burden, and enable more meaningful patient–clinician interaction.

At the same time, patients express concerns related to loss of human connection, lack of transparency or explainability, inconsistent performance across care settings, privacy and secondary use of health data, and accountability when AI contributes to clinical decisions. Evidence suggests patient trust increases when AI tools are transparent, grounded in high-quality clinical data, evaluated in real-world settings, and deployed as decision support rather than decision-makers.